

Challenge 4

A large organization is migrating its on-premises infrastructure to the AWS cloud.

The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

Objectives:

- Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.
- Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.

AWS Transit Gateway + VPC Endpoint —

1. Create 4 VPCs
 - VPC-1 → 10.10.0.0/16
 - VPC-2 → 10.20.0.0/16
 - VPC-3 → 10.30.0.0/16
 - VPC-4 → 10.40.0.0/16
2. Create 1 subnet in each VPC
 - VPC-1 → 10.10.1.0/24
 - VPC-2 → 10.20.1.0/24
 - VPC-3 → 10.30.1.0/24
 - VPC-4 → 10.40.1.0/24
3. Create Transit Gateway
 - Name: Company-TGW
4. Create 4 TGW attachments
 - Attach VPC-1, VPC-2, VPC-3, VPC-4
 - Wait for Available.
5. Configure VPC route tables
 - Add other VPC CIDRs → Transit Gateway
6. Create VPC Endpoint
 - VPC → VPC Endpoints → Create
 - Service: S3
 - Type: Gateway
 - Select VPC-1
 - Select its route table
 - Create.
7. Verify
 - 4 VPCs ✓
 - 4 subnets ✓
 - TGW + 4 attachments ✓

- Routes 
- S3 Gateway Endpoint → Available 

Transit gateway attachments (4) [Info](#)

☐	Name ↗	Transit gateway attachment ID	Transit gateway ID	State	Resource t...
☐	TGW-VPC4-DB	tgw-attach-050d2d8f0039f0eb7	tgw-000876579c3fbec2b	✓ Available	VPC
☐	Company-TGW	tgw-attach-061b2632cf5cee85e	tgw-000876579c3fbec2b	✓ Available	VPC
☐	TGW-VPC2-IT	tgw-attach-0906a01b02512d866	tgw-000876579c3fbec2b	✓ Available	VPC
☐	TGW-VPC3-APP	tgw-attach-0aed447dceaa0465f	tgw-000876579c3fbec2b	✓ Available	VPC

Transit gateway route tables (1/1) [Info](#)

☒	Name ↗	Transit gateway route table ID	Transit gateway ID	State	Default association route ta
☒		tgw-rtb-0da833def0ae32ad1	tgw-000876579c3fbec2b	✓ Available	Yes

Transit gateway route tables: tgw-rtb-0da833def0ae32ad1

Associations (4) [Info](#)

☐	Attachment ID	Attachment name	Resource type	Resource ID	State
☐	tgw-attach-0aed447dceaa0465f	TGW-VPC3-APP	VPC	vpc-0738fdb5ecb714289	✓ Associated
☐	tgw-attach-050d2d8f0039f0eb7	TGW-VPC4-DB	VPC	vpc-0175769590e286253	✓ Associated
☐	tgw-attach-0906a01b02512d866	TGW-VPC2-IT	VPC	vpc-07f29fb51f95c57d0	✓ Associated
☐	tgw-attach-061b2632cf5cee85e	Company-TGW	VPC	vpc-0d1da8ed93f42a9ac	✓ Associated

VPC dashboard [←](#)

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

rtb-0d3fb131db3cc3618 [Actions](#)

Details [Info](#)

Route table ID rtb-0d3fb131db3cc3618	Main ☒ Yes	Explicit subnet associations -	Edge associations -
VPC vpc-0d1da8ed93f42a9ac VPC-1-HR	Owner ID 063298174631		

Routes | Subnet associations | Edge associations | Route propagation | Tags | Related resources

Routes (6)

Destination	Target	Status	Propagated	Route Origin
pl-c3aa4faa	vpce-0989d217fe6244315	✓ Active	No	Create Route
10.0.0.0/16	local	✓ Active	No	Create Route Table
10.10.0.0/16	local	✓ Active	No	Create Route Table
10.20.0.0/16	tgw-000876579c3fbec2b	✓ Active	No	Create Route
10.30.0.0/16	tgw-000876579c3fbec2b	✓ Active	No	Create Route
10.40.0.0/16	tgw-000876579c3fbec2b	✓ Active	No	Create Route

VPC

> Route tables

Filter by VPC:

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PrivateLink and Lattice

Route tables (1/5)

Info

Last updated 6 minutes ago

Actions

Create route table

Find route tables by attribute or tag

Name

Route table ID

Explicit subnet associ...

Edge associations

Main

VPC

-

rtb-0e6f548f5c582e4d9

-

-

Yes

vpc-04069a29t

☒

-

rtb-0970e4ad506c57f1f

-

-

Yes

vpc-01757695t

-

rtb-0d3fb131db3cc3618

-

-

Yes

vpc-0d1da8edf

-

rtb-035a0a86660a2b4ea

-

-

Yes

vpc-0738fdb5e

rtb-0970e4ad506c57f1f

Details

Routes

Subnet associations

Edge associations

Route propagation

Tags

Related resources

Routes (5)

Filter routes

Destination

Target

Status

Propagated

Route Origin

10.3.0.0/16

local

Active

No

Create Route Table

10.10.0.0/16

tgw-000876579c3fbec2b

Active

No

Create Route

10.20.0.0/16

tgw-000876579c3fbec2b

Active

No

Create Route

10.30.0.0/16

tgw-000876579c3fbec2b

Active

No

Create Route

10.40.0.0/16

local

Active

No

Create Route Table

VPC

> Endpoints

vpce-0989d217fe6244315

Filter by VPC:

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vpce-0989d217fe6244315

Details

Route tables

Policy

Tags

Endpoint ID

vpce-0989d217fe6244315

Status

Available

Creation time

Monday, 21 September 2026 at 19:30:47 GMT+5:30

Endpoint type

Gateway

VPC ID

vpc-0d1da8ed93f42a9ac (VPC-1-HR)

Status message

-

Service name

com.amazonaws.eu-north-1.s3

Private DNS names enabled

No

DNS record IP type

service-defined

IP address type

ipv4

Service region

eu-north-1

Private DNS specified domains

-

Route tables (1)

Search

Route Table ID

Main

Associated Id

rtb-0d3fb131db3cc3618

Yes

subnet-04386451e997accb (VPC1-Su...

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-transit-gateways --output table
```

DescribeTransitGateways	
TransitGateways	
CreationTime	2026-09-21T13:30:34+00:00
Description	Central TGW for all VPCs
OwnerId	063298174631
State	available
TransitGatewayArn	arn:aws:ec2:eu-north-1:063298174631:transit-gateway/tgw-000876579c3fbec2b
TransitGatewayId	tgw-000876579c3fbec2b
Options	
AmazonSideAsn	64512
AssociationDefaultRouteTableId	tgw-rtb-0da833def0ae32ad1
AutoAcceptSharedAttachments	disable
DefaultRouteTableAssociation	enable
DefaultRouteTablePropagation	enable
DnsSupport	enable
MulticastSupport	disable
PropagationDefaultRouteTableId	tgw-rtb-0da833def0ae32ad1
SecurityGroupReferencingSupport	disable
VpnEcmpSupport	enable
EncryptionSupport	
EncryptionState	disabled
Tags	
Key	Name
Value	Company-TGW

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
```

```
vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-transit-gateway-vpc-attachments --output table
```

DescribeTransitGatewayVpcAttachments		
TransitGatewayVpcAttachments		
CreationTime	2026-09-21T13:34:30+00:00	
State	available	
TransitGatewayAttachmentId	tgw-attach-050d2d8f0039f0eb7	
TransitGatewayId	tgw-000876579c3fbec2b	
VpcId	vpc-0175769590e286253	
VpcOwnerId	063298174631	
options		
ApplianceModeSupport	disable	
DnsSupport	enable	
Ipv6Support	disable	
SecurityGroupReferencingSupport	enable	
SubnetIds		
subnet-0396a1352629adff2		
Tags		
Key	Name	
Value	TGW-VPC4-DB	
TransitGatewayVpcAttachments		
CreationTime	2026-09-21T13:32:46+00:00	
State	available	
TransitGatewayAttachmentId	tgw-attach-061b2632cf5cee85e	
TransitGatewayId	tgw-000876579c3fbec2b	
VpcId	vpc-0d1da8ed93f42a9ac	
VpcOwnerId	063298174631	
options		
ApplianceModeSupport	disable	
DnsSupport	enable	
Ipv6Support	disable	
SecurityGroupReferencingSupport	enable	
SubnetIds		
subnet-04386451e997accb		
Tags		
Key	Name	
Value	Company-TGW	
TransitGatewayVpcAttachments		
CreationTime	2026-09-21T13:33:26+00:00	
State	available	
TransitGatewayAttachmentId	tgw-attach-0906a01b02512d866	
TransitGatewayId	tgw-000876579c3fbec2b	
VpcId	vpc-07f29fb51f95c57d0	
VpcOwnerId	063298174631	
options		
ApplianceModeSupport	disable	
DnsSupport	enable	
Ipv6Support	disable	
SecurityGroupReferencingSupport	enable	
SubnetIds		
subnet-09489d24abd0901cb		
Tags		
Key	Name	
Value	TGW-VPC2-IT	
TransitGatewayVpcAttachments		
CreationTime	2026-09-21T13:34:00+00:00	
State	available	
TransitGatewayAttachmentId	tgw-attach-0aed447dceaa0465f	
TransitGatewayId	tgw-000876579c3fbec2b	
VpcId	vpc-0738fdb5ecb714289	
VpcOwnerId	063298174631	
options		
ApplianceModeSupport	disable	
DnsSupport	enable	
Ipv6Support	disable	
SecurityGroupReferencingSupport	enable	
SubnetIds		
subnet-015f877c1c9174454		
Tags		
Key	Name	
Value	TGW-VPC3-APP	

```
vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
$ aws ec2 describe-vpc-endpoints --output table
```

DescribeVpcEndpoints	
VpcEndpoints	
CreationTimestamp	2026-09-21T14:00:47+00:00
IpAddressType	ipv4
OwnerId	063298174631
PolicyDocument	{"Version": "2008-10-17", "Statement": [{"Effect": "Allow", "Principal": "*", "Action": "*", "Resource": "*}]}
PrivateDnsEnabled	False
RequesterManaged	False
ServiceName	com.amazonaws.eu-north-1.s3
ServiceRegion	eu-north-1
State	available
VpcEndpointId	vpce-0989d217fe6244315
VpcEndpointType	Gateway
VpcId	vpc-0d1da8ed93f42a9ac
DnsOptions	
DnsRecordIpType	service-defined
RouteTableIds	
rtb-0d3fb131db3cc3618	

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~ (master)
```